Viability-Preserving Passive Torque Control for Mobile Base Manipulator

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Introduction/Abstract

- Implemented a **viability-preserving** torque control framework [1] for a 7-DOF robot arm on a 3-DOF omni-directional mobile base to ensure safe, collision-free tracking
- Enforces constraints on joint states to avoid self-collisions, joint limits, and obstacles through **precomputed safe sets** and instantaneous quadratic programming-based control.
- We draw comparison with a **CBF based Instantaneous QP** control method with the same constraint objectives.

Methods

Viable Set: For any state $(q(0), \dot{q}(0))$ in viable set, there exists at least one control input that keeps $(q, \dot{q})$ in the feasible set $\mathcal{F}$.

$$\exists(\ddot{q}_t)_{t=0}^{\infty} : \big(q(t), \dot{q}(t)\big) \in \mathcal{F}, \forall t \geq 0$$

Self-collision-free viability: $V_{sca} \coloneqq \{(q, \dot{q}) | \Gamma(\mathrm{q}, \dot{\mathrm{q}}) > 0\}$

Learnt $\Gamma(q, \dot{q})$ using **Transformer** NN architecture

External-collision-free viability: $V_{eca} \coloneqq \{(q, \dot{q}) | S_v(p, q, \dot{q}) > 0\}$

Learnt $S_v(q, \dot{q})$ using **Bernstein Polynomial basis functions**

Joint limits viability: $\boldsymbol{V_{jnt} \coloneqq \{\ddot{q} : \ddot{q}_{lb} < \ddot{q} < \ddot{q}_{ub}\}}$

Optimization Problem [1]:

$$\min_{\tau} 0.5\left\|\hat{J}(q)^{-\mathrm{T}}\tau - F_c(x)\right\|^2 + \alpha\|\tau\|^2$$

- **Self-Collision Constraint** $\Delta\Gamma(q, \dot{q}) > 0$ gives

$$\boldsymbol{g_{se}M^{-1}\tau \geq g_{se}M^{-1}(C\dot{q} + G) - b_s}$$
$$g_{se} = f\left(\nabla_q\Gamma, \nabla_{\dot{q}}\Gamma\right),\ b_s = g(\nabla_q\Gamma)$$

- **External-Collision Constraint** $\Delta S_v(q, \dot{q}) > 0$ gives

$$\boldsymbol{g_{ee}M^{-1}\tau \geq g_{ee}M^{-1}(C\dot{q} + G) - b_e}$$
$$g_{se} = f\left(\nabla_q \mathrm{S_v}, \nabla_{\dot{q}} \mathrm{S_v}\right),\ b_s = g(\nabla_q S_v)$$

- **Joint position, velocity and acceleration limits** set through analytical formulas described in [1]

Methods

Baseline method: Control Barrier Functions in [2].
Corresponding three constraints:

1. Self-Collision Avoidance

$$\nabla_q h_{SCA}(q)\ddot{q} \geq b_{SCA}(q)$$
$$h_{SCA}(q) = \Gamma_{SCA}(q) - \epsilon_{SCA}$$
$$b_{SCA}(q) = -k_1 h_{SCA}(q) - k_2 \nabla_q h_{SCA}(q)\dot{q} - \dot{q}^T \nabla_q^2 h_{SCA}(q)\dot{q}$$

2. External Collision Avoidance

$$\nabla_q h_{ECA}^{(i)}(q)\ddot{q} \geq b_{ECA}^{(i)}(q)$$
$$h_{ECA}^{(i)}(q) = \Gamma_{SDF}^{(i)}(q, x_0) - \epsilon_{SDF}$$
$$b_{ECA}^{(i)}(q) = -k_1 h_{ECA}^{(i)}(q) - k_2 \nabla_q h_{ECA}^{(i)}(q)\dot{q} - \dot{q}^T \nabla_q^2 h_{ECA}^{(i)}(q)\dot{q}$$

3. Joint Limit Avoidance

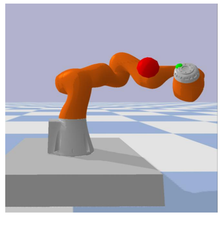
Key differences:

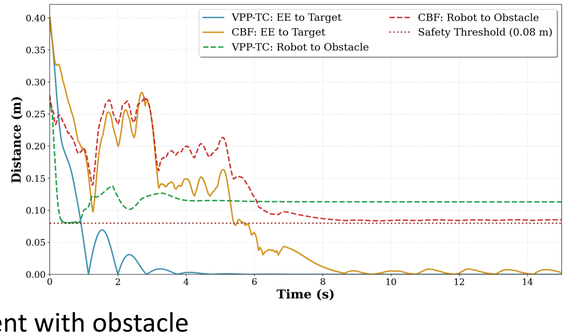
- Safety Formulation – Hessian based constraints
- Nature of Safety Guarantee – Predictive vs Reactive
- Role of Learning Models

Experiments and Results

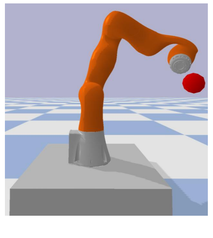
Fixed Base Experiments:

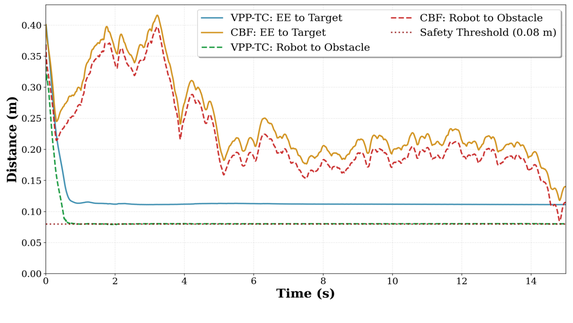
1. Obstacle before target point





2. Target coincident with obstacle

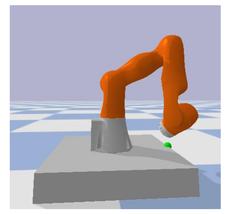


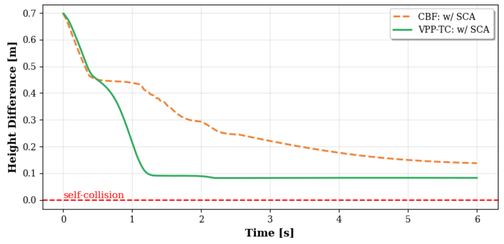


Experiments and Result

Fixed Base Experiments:

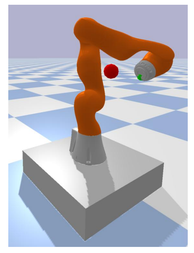
3. Target inside robot link (Self-Collision)

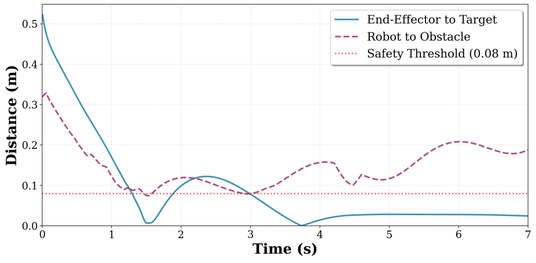




Mobile Base Experiments:

1. All three constraints activated





Observations/Conclusions

- **Control Loop Rate**:
  VPP-TC is considerably faster. CBF-TC requires hessian Calculation

VPP-TC	CBF-TC
110 Hz	47 Hz

- **Numerical Stability:** VPP-TC is stabler and converges faster than CBF-TC. Because of numerical instability, CBF-TC jitters during control, as evident in graphs
- **Future work:**

1. Improve the current mobile base strategy for more complex and dynamic tasks
2. Use in reward shaping for RL controllers

References

[1] Z. Zhang, Y. Wang, N. Figueroa, et al., "Viability-Preserving Passive Torque Control," under review, 2024.
[2] Z. Zhang, T. Li and N. Figueroa, "Constrained Passive Interaction Control: Leveraging Passivity and Safety for Robot Manipulators," 2024 IEEE International Conference on Robotics and Automation (ICRA), 2024.